

STAT 6611: Theory & Application of Linear Models

1 Preliminaries: Generalized Inverses/Projection Matrices

Theorem 1.1. *For any $m \times n$ matrix A of rank $r \geq 1$, there exist nonsingular matrices P (size $m \times m$) and Q (size $n \times n$) such that*

$$PAQ = \begin{bmatrix} D_{r \times r} & 0_{r \times n-r} \\ 0_{m-r \times r} & 0_{m-r \times n-r} \end{bmatrix},$$

where $D \equiv D_{r \times r}$ is a nonsingular $r \times r$ diagonal matrix.

The $n \times m$ matrix

$$G = Q \begin{bmatrix} D^{-1} & F \\ H & B \end{bmatrix} P$$

satisfies $AGA = A$, where F, H , and B are any matrices of appropriate dimensions.

Proof. Note that

$$PAQ = \begin{bmatrix} D & 0 \\ 0 & 0 \end{bmatrix} \iff A = P^{-1} \begin{bmatrix} D & 0 \\ 0 & 0 \end{bmatrix} Q^{-1}.$$

Let

$$A = \begin{bmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & \vdots & \vdots \\ a_{m1} & \cdots & a_{mn} \end{bmatrix} = (\underline{a}_1, \dots, \underline{a}_n)$$

represent an $m \times n$ matrix of rank r ($r \geq 1$).

Take K_1 to be an $m \times r$ matrix whose columns are r linearly independent columns of A , say

$$K_1 = (\underline{a}_{s_1}, \dots, \underline{a}_{s_r}).$$

Every column of A must then be a linear combination of the columns of K_1 .

In other words, for some $t_{11}, \dots, t_{r1}, \dots, t_{1n}, \dots, t_{rn}$,

$$\begin{aligned} \underline{a}_1 &= t_{11}\underline{a}_{s_1} + \dots + t_{r1}\underline{a}_{s_r} = (\underline{a}_{s_1}, \dots, \underline{a}_{s_r}) \begin{pmatrix} t_{11} \\ \vdots \\ t_{r1} \end{pmatrix} = K_1 \underline{t}_1, \\ &\vdots \\ \underline{a}_n &= t_{1n}\underline{a}_{s_1} + \dots + t_{rn}\underline{a}_{s_r} = (\underline{a}_{s_1}, \dots, \underline{a}_{s_r}) \begin{pmatrix} t_{1n} \\ \vdots \\ t_{rn} \end{pmatrix} = K_1 \underline{t}_n. \end{aligned}$$

Hence,

$$A = (\underline{a}_1, \dots, \underline{a}_n) = (K_1 \underline{t}_1, \dots, K_1 \underline{t}_n) = K_1 (\underline{t}_1, \dots, \underline{t}_n) = K_1 T_1$$

for some $r \times n$ matrix $T_1 = (\underline{t}_1, \dots, \underline{t}_n)$.

It turns out that $\text{rank}(T_1) = r$.

Next, take K to be an $m \times m$ nonsingular matrix and T to be an $n \times n$ nonsingular matrix of the forms

$$K = (K_1, K_2) \quad T = \begin{pmatrix} T_1 \\ T_2 \end{pmatrix}.$$

Observe that

$$A = K_1 T_1 = (K_1, 0) \begin{pmatrix} T_1 \\ T_2 \end{pmatrix} = (K_1, K_2) \begin{bmatrix} I_r & 0 \\ 0 & 0 \end{bmatrix} \begin{pmatrix} T_1 \\ T_2 \end{pmatrix} = K \begin{bmatrix} I_r & 0 \\ 0 & 0 \end{bmatrix} T,$$

where I_r is the $r \times r$ identity matrix.

Hence,

$$K^{-1} A T^{-1} = \begin{bmatrix} I_r & 0 \\ 0 & 0 \end{bmatrix},$$

and $P = K^{-1}$, $Q = T^{-1}$, and $D = I_r$ satisfy the requirements.

Finally, given the existence of P, Q , and D , we have

$$\begin{aligned} AGA &= P^{-1} \begin{bmatrix} D & 0 \\ 0 & 0 \end{bmatrix} Q^{-1} Q \begin{bmatrix} D^{-1} & F \\ H & B \end{bmatrix} P P^{-1} \begin{bmatrix} D & 0 \\ 0 & 0 \end{bmatrix} Q^{-1} \\ &= P^{-1} \begin{bmatrix} D & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} D^{-1} & F \\ H & B \end{bmatrix} \begin{bmatrix} D & 0 \\ 0 & 0 \end{bmatrix} Q^{-1} \\ &= P^{-1} \begin{bmatrix} I_r & DF \\ 0 & 0 \end{bmatrix} \begin{bmatrix} D & 0 \\ 0 & 0 \end{bmatrix} Q^{-1} = P^{-1} \begin{bmatrix} D & 0 \\ 0 & 0 \end{bmatrix} Q^{-1} = A. \end{aligned}$$

□

Definition. A generalized inverse of a matrix A is any matrix G that satisfies $AGA = A$.

Notation: For any matrix A , we use A^- to denote an arbitrary generalized inverse of A .

Theorem 1.1 indicates that:

- any matrix A has a generalized inverse
- if $A = 0$ is the $m \times n$ zero matrix, then any $n \times m$ matrix G is a generalized inverse of A .
- A has infinitely many generalized inverses unless A is nonsingular
- if A is nonsingular, it has a unique generalized inverse, which is A^{-1} .
- Also, Theorem 1.1 suggests an algorithm for computing a generalized inverse.
- Note in Theorem 1.1 that values of P, Q, D are not unique.

Definition. The equations $A\underline{x} = \underline{y}$ are said to be consistent if they have at least one solution.

Theorem 1.2. *For any matrix A , a matrix G is a generalized inverse of A if and only if $\underline{x} = G\underline{y}$ is a solution to the equations $A\underline{x} = \underline{y}$ for every vector $\underline{y}$ for which the equations are consistent.*

Remark. Let A be an $m \times n$ matrix and X an $n \times p$ matrix. Then, $Y = AX$ is an $m \times p$ matrix. Let $\underline{x}_k$ be the k th column of X and $\underline{y}_k$ be the k th column of Y . Then,

$$AX = Y \iff A(\underline{x}_1, \dots, \underline{x}_p) = (\underline{y}_1, \dots, \underline{y}_p) \iff A\underline{x}_k = \underline{y}_k \text{ for all } k.$$

Thus, Theorem 1.2 can be extended to the matrix equation $AX = Y$.

That is, for any matrix A , a matrix G is a generalized inverse of A if and only if $X = GY$ is a solution to the matrix equation $AX = Y$ for every matrix Y for which the equation is consistent.

Theorem 1.3. *All solutions to a consistent matrix equation $AX = Y$ are generated by putting*

$$X = A^-Y + (I - A^-A)Z$$

and letting Z range over the space of all possible matrices (of appropriate dimensions), where I denotes the identity matrix (of appropriate dimensions).

Proof. Based on the remark after Theorem 1.2, we have

$$A[A^-Y + (I - A^-A)Z] = A(A^-Y) + A(I - A^-A)Z = A(A^-Y) + (A - AA^-A)Z = A(A^-Y) = Y,$$

for any Z (of appropriate dimensions).

Conversely, for any solution X^* to $AX = Y$,

□

Lemma 1.1. *A matrix G is a generalized inverse of a matrix A if and only if G' is a generalized inverse of A' , where A' is the transpose of A .*

Corollary. *If A is a symmetric matrix, then $(A^-)'$ is a generalized inverse of A . In particular, for any matrix X , $[(X'X)^-]'$ is a generalized inverse of $X'X$.*

Remark. A generalized inverse of a symmetric matrix A can be nonsymmetric.

Lemma 1.2. *For any matrix X , $\text{tr}(X'X) = 0$ implies $X = 0$.*

Proof. Let $\underline{x}_j$ be the j th column of X . Note that

$$X'X = (\underline{x}_1, \dots, \underline{x}_n)'(\underline{x}_1, \dots, \underline{x}_n) = \begin{pmatrix} \underline{x}_1' \\ \vdots \\ \underline{x}_n' \end{pmatrix} (\underline{x}_1, \dots, \underline{x}_n) = \begin{bmatrix} \underline{x}_1'\underline{x}_1 & \cdots & \underline{x}_1'\underline{x}_n \\ \vdots & \vdots & \vdots \\ \underline{x}_n'\underline{x}_1 & \cdots & \underline{x}_n'\underline{x}_n \end{bmatrix}.$$

Lemma 1.3. *For any matrix X and for any matrices P and Q of appropriate dimensions, we have*

- (i) $PX'X = QX'X$ implies $PX' = QX'$ and
- (ii) $X'XP = X'XQ$ implies $XP = XQ$.

Proof of (i): $PX'X = QX'X$ implies that

$$0 = (PX'X - QX'X)(P - Q)' = (XP' - XQ')'X(P' - Q') = (XP' - XQ')'(XP' - XQ').$$

Lemma 1.4. *For any matrix X , $X'X(X'X)^-X' = X'$, that is, $X(X'X)^-$ is a generalized inverse of X' .*

Corollary 1. (i) $X'X[(X'X)^-]'X' = X'$, (ii) $X(X'X)^-X'X = X$, and (iii) $X[(X'X)^-]'X'X = X$.

Corollary 2. Corresponding to any matrix X , there exists a matrix B satisfying $X'XB = X'$.

Lemma 1.5. *Let X be any matrix and let B_1 and B_2 be any two matrices that satisfy $X'XB = X'$. Then, $XB_1 = XB_2$.*

Corollary 1. Let X be any matrix and let B be any matrix that satisfies $X'XB = X'$. Then, $XBX = B'X'X = X$.

Corollary 2. Let X be any matrix and let G_1 and G_2 be any two generalized inverses of $X'X$. Then, $XG_1X' = XG_2X'$.

Corollary 3. For any matrix X , $X(X'X)^-X' = X[(X'X)^-]'X'$, that is, $X(X'X)^-X'$ is symmetric.

Corollary 4. Let X be any matrix and let B be any matrix that satisfies $X'XB = X'$. Then, the matrix XB is symmetric.

Corollary 5. Let $P_X = XB$ where B is a solution to $X'XB = X'$. Then, P_X has the following properties:

(1) P_X is invariant to the choice of solution B .

(2) P_X is symmetric.

(3) $P_X = X(X'X)^{-1}X'$.

(4) $P_X X = X$.

(5) P_X is idempotent, that is, $P_X^2 = P_X$

Definition. If $P_X = XB$ where B is a solution to $X'XB = X'$, then P_X is said to be an orthogonal projection matrix.

Definition. The column space of an $m \times n$ matrix A consists of all $m \times 1$ vectors spanned by the columns of A , that is, a vector $\underline{c}$ belongs to the column space of A if and only if there exists a vector $\underline{b}$ for which $\underline{c} = A\underline{b}$.

Similarly, the row space of A consists of all $1 \times n$ vectors spanned by the rows of A .

Notation. We use $\mathcal{C}(A)$ to denote the column space of A and use $\mathcal{R}(A)$ to denote the row space of A .

Lemma 1.6. (i) $\mathcal{C}(B)$ is a subspace of $\mathcal{C}(A)$ if and only if $B = AC$ for some matrix C . (ii) $\mathcal{R}(B)$ is a subspace of $\mathcal{R}(A)$ if and only if $B = CA$ for some matrix C .

Various special cases of the following two theorems are very useful in linear-model theory and methodology.

Theorem 1.4. *Let $M = \begin{bmatrix} A & C \\ D & B \end{bmatrix}$ be a partitioned matrix.*

(i) *If $\mathcal{C}(C)$ is a subspace of $\mathcal{C}(A)$ and $\mathcal{R}(D)$ is a subspace of $\mathcal{R}(A)$, then the partitioned matrix*

$$\begin{bmatrix} A^- + A^-CQ^-DA^- & -A^-CQ^- \\ -Q^-DA^- & Q^- \end{bmatrix}$$

is a generalized inverse of M , where $Q = B - DA^-C$.

(ii) *If $\mathcal{C}(D)$ is a subspace of $\mathcal{C}(B)$ and $\mathcal{R}(C)$ is a subspace of $\mathcal{R}(B)$, then the partitioned matrix*

$$\begin{bmatrix} P^- & -P^-CB^- \\ -B^-DP^- & B^- + B^-DP^-CB^- \end{bmatrix}$$

is a generalized inverse of M , where $P = A - CB^-D$.

Theorem 1.5. *Let $W = R + D$, where $\mathcal{C}(D)$ is a subspace of $\mathcal{C}(R)$ and $\mathcal{R}(D)$ is a subspace of $\mathcal{R}(R)$ (as would be the case, for example, if R were nonsingular). Take S, T , and U to be any three matrices such that*

$$D = STU.$$

Then, the matrix

$$R^- - R^-ST(T + TUR^-ST)^-TUR^-$$

is a generalized inverse of W .